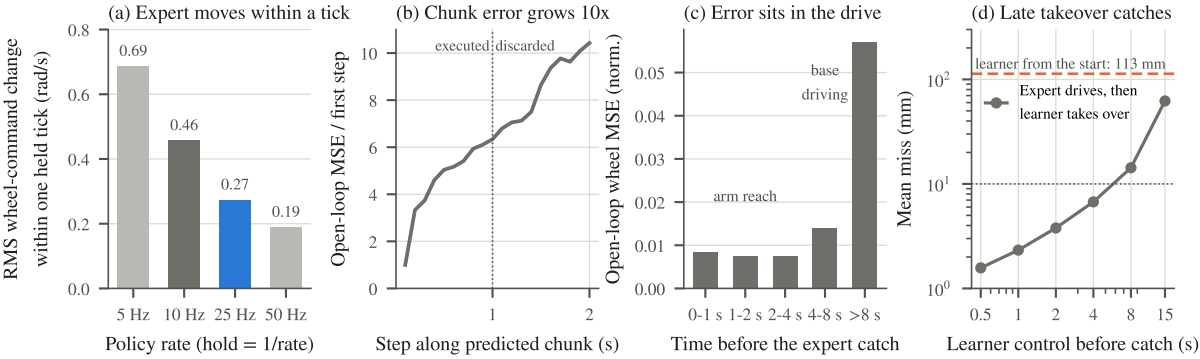
(a) Expert moves within a tick
RMS wheel-command change within one held tick (rad/s)
0.0
0.2
0.4
0.6
0.8
0.69
0.46
0.27
0.19
5 Hz
10 Hz
25 Hz
50 Hz
Policy rate (hold = 1/rate)
(b) Chunk error grows 10x
Open-loop MSE / first step
executed
discarded
0
2
4
6
8
10
1
2
Step along predicted chunk (s)
(c) Error sits in the drive
Open-loop wheel MSE (norm.)
0.00
0.01
0.02
0.03
0.04
0.05
base
driving
arm reach
0-1 s
1-2 s
2-4 s
4-8 s
>8 s
Time before the expert catch
(d) Late takeover catches
Mean miss (mm)
learner from the start: 113 mm
Expert drives, then
learner takes over
10^0
10^1
10^2
0.5
1
2
4
8
15
Learner control before catch (s)